MEASURING DAMAGES UNDER THE CONVENTION ON INTERNATIONAL LIABILITY FOR DAMAGE CAUSED BY SPACE OBJECTS +

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"Contrary to some early scientific expectation that fragments and parts of objects placed in orbit or otherwise sent to outer space would burn up in the atmosphere prior to reaching earth unless designed and made for reentry, some have fallen back to earth as solid pieces. Even though the incidence of damage has been low. . . as the number of objects put into outer space increases, the probability of damage from falling debris might increase."[1]

Although the semantics belie the potential tragedy which can result from such falling debris, the validity of the above quoted observation was irrevocably established on January 21, 1978 when the nuclear powered Soviet satellite fell from orbit into the Canadian forests.[2] Although Cosmos 954 grounded in a relatively unpopulated area, it could just as readily have impacted in a major urban area with the attendant injury and destruction to lives and property.

It has long been recognized that such space flights and exploration pose a risk not only for direct participants, but also for all those who remain earthbound.[3] This risk of injury has in turn produced a multitude of unresolved questions concerning liability. For example, who should bear liability for injury? Should that liability be dependent upon fault notions or be based upon strict liability? What interests should be protected? What remedies ought to be available to protect those interests? If the available remedies

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[1]U.S. Sen., Report, on The Convention on International Liability for Damage Caused by Space Objects Launched into Outer Space - Analysis and Background Data, 92d Cong., 2d Sess. 6 (1972). [hereinafter cited as Senate Rept.]

[2]Satchell, *Can Our Space Scientists Keep Skylab Aloft,* Akron Beacon Journal, May 7, 1978, at 4 (Parade Magazine).

[3]"Since injury or damage might result from a launching, flight and return to earth of various kinds of space vehicles or parts thereof, a number of problems exist with respect to defining and delimiting liability of the launching State and other States associated with it in the space activity *causing injury or damage.*" Report of the *Ad Hoc* committee on the Peaceful Uses of Outer Space of its 14th Sess., UN Doc. A/4141, July 14, 1959. (Emphasis added).

include damages, then how should such awards be measured? And finally, should damage liability be limited in amount?[4]

Some of these questions were initially addressed in the Outer Space Treaty.[5] That treaty established the general principle that the launching state has a liability for the injury caused to persons by its launched objects.[6] One progeny of the Outer Space Treaty, the Convention on International Liability for Damage Caused by Space Objects[7] was conceived to flesh out this principle: "to establish international rules and procedures concerning liability for damage caused by the launching of objects into outer space and to insure, in particular, the prompt and equitable compensation for damage."[8] Its legislative history evidences further an intent that the Liability Convention "contain provisions which would insure the payment of a full measure of compensation to victims and effective procedures which would lead to the prompt and *equitable settlement of claims.*"[9]

Although it is not the purpose of this paper to detail the Liability Convention's procedure for processing claims, it is important to note that the process contemplates settlement through diplomatic negotiations or via a Claims Commission. If diplomatic negotiations fail to produce a settlement, then the party States appoint a Claims Commission to process that claim.[10] Whether settlement is reached through diplomatic negotiations or as a result of the Claims Commission's determinations, the final award must achieve two goals specifically fixed by the Liability Convention: (1) compensation must be determined "in accordance with international law and the principles of justice and equity";[11] and (2) the damage award must "restore the person. . . to the condition which would have existed if the damage had not occurred."[12] The issue of applicable

[4] *See*, Senate Rept., *supra* n. 1 at 7.

[5] Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space Including the Moon and Other Celestial Bodies, Jan. 27, 1967 18 UST 2410, TIAS No. 6347.

[6] *Id.* at Art. VII. It has stated that "this principle . . . acknowledges the following varieties of the connection between the cause of damage and the State responsible for *compensation:* (1)launching (2)procuring the launching; and (3)putting the territory or facilities at the disposal for launching." Senate Rept. *supra* n. 1 at 8. (Emphasis added).

[7] Convention on International Liability for Damage Caused By Space Objects March 29, 1972 (1973) 24 U.S.T. 2389, T.I.A.S. 7762 (hereinafter cited as Liability Convention).

[8] G.A. Res. 2601B (XXIV) of Dec. 16, 1969.

[9] G.A. Res. 2733B (XXV) of Dec. 16, 1970. (Emphasis added). This principle was reaffirmed in G.A. Res. 2777 (XXVI) of Nov. 29, 1971.

[10] Art. XIV, Liability Convention.

[11] *Id.*, Art. XII

[12] *Ibid.*

law (that of the claimant State or the State with potential liability) was apparently a major hurdle to reaching a final draft of the Liability Convention. "Since no agreement was reached on the applicability of the national law of a single State, accord was finally reached on the general principles. . . that the compensation for damage 'shall be determined in accordance with international law and the principles of justice and equity, . . .'"[13] Even though the Liability Convention omits a rigid rule for resolving the question of applicable law, the statement of the two remedial goals may have mitigated the possible adverse consequences of this omission. By noting that compensation is to be determined by applying the principles of justice and equity, the drafters of the Convention have dictated that regardless of the applicable law ultimately selected during the award-fixing procedure, application of that law must achieve equity and justice.

The phrase "justice and equity" makes clear that the term "equity" is used in its popular sense to signify moral justice and not in an Anglo-American legal sense.[14] As one scholar has noted, in this "general juristic sense, equity means the power to meet the moral standards of justice in a particular case by a tribunal having discretion to mitigate the rigidity of the application of strict rules of law so as to adapt relief to the circumstances of the particular case."[15] Indeed, "the need of principles of discretion in applying legal precepts and remedies, of dispensing from strict application of rules under special circumstances or in unusual relations. . . is even more imperative under the complicated and involved relations and circumstances of an increasingly crowded world."[16]

What standard of compensation will do justice to an injured party in the event of injury from falling space objects? At least that measure which will fulfill the Convention's second remedial goal: a measure of relief which will restore the injured persons "to the condition which would have existed if the damage had not occurred."[17] Perhaps in recognition that the most likely injury resulting from a falling space object cannot be remedied by specific restoration of the *status quo ante,* the drafters of the Liability Convention elected instead substitutionary relief. That is, compensating the

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[13] *Senate Rept., supra* n. 1 at 19.

[14] *See,* de Funiak, Handbook on Modern Equity § 1 (2 ed. 1956).

[15] McClintock, McClintock on Equity 1 (2 ed. 1948).

[16] Neuman, Equity in Law: A Comparative Study 10 (1961).

[17] Art. XII, Liability Convention.

injured person for the injury to his person or property.[18] A third remedy not addressed in the Convention is the award of nominal damages in declaration of an injured person's rights.[19] The Liability Convention's silence as to nominal damages need not preclude their availability. The goals of equity and justice could permit an award-making tribunal to determine that the rights of an injured party had been wrongfully infringed by the space object's launching State.[20] The allocation of fault through such a declaration of rights would serve little purpose, however, in light of another provision of the Liability Convention.

That provision states that "[a] launching State shall be absolutely liable to pay compensation for damages caused by its space object on the surface of the earth. . ."[21] Neither negligence nor other standards of fault need be proven to produce a remedial recovery. All that need be established is the presence of the requisite damages. The Convention defines "damage" to mean "loss of life, personal injury or impairment of health, or loss of or damage to property of States or persons, natural or juridical, or property of international intergovernmental organizations."[22] In order to explore each of these several categories of "damage" to determine what measure of recovery would be sufficient to do justice and to also restore the injured person to the condition which would have existed but for the injury, assume that the space object landed on real property owned by Mr. Harry Smith. Both Mr. Smith and Mr. Bob Jones were standing on that property at the moment of impact by the space object. Both Mr. Smith and Mr. Jones sustained personal injuries as a result of the impact. Both were hospitalized and received medical treatment over the next several weeks. Mr. Smith eventually died from his injuries, survived by his wife and three minor children. Further assume that the launching state has absolute liability in this situation. But who is the proper claimant(s)? Had Mr. Smith lived, he could have asserted a compensable claim for his personal

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[18] "[T]he law of torts attempts primarily to put an injured person in a position as nearly as possible equivalent to his position prior to the tort. The law was able to do this only in varying degrees dependent upon the nature of the harm. Thus, where the plaintiff has been harmed in body or mind, money damages are no equivalent but are given to compensate the plaintiff for the pain or for the deterioration of the bodily structure. . .in other situations, as where there has been harm to earning capacity, the law can indemnify the plaintiff for pecuniary loss, such indemnity not being the exact equivalent but one which approximates the pecuniary harm which the injured person has suffered or is likely to suffer in the future." Restatement of Torts, § 901, comment *a*.

[19] *Id.* at §907. *See also,* York & Bauman, Remedies, Cases and Materials 2-3 (2d ed. 1973).

[20] The United States Supreme Court has very recently reaffirmed the importance of nominal damages as a vehicle for the declaration of rights of aggrieved persons. *Carey v. Piphus,* 98 S. Ct. 1042, (1978). The court also noted that an award not to exceed one dollar would accomplish this remedial goal. *Id.* at p. 1054. *See also,* Dobbs, Law of Remedies, § 3.8, (1973); McCormick, Law of Damages, §§ 20-22 (1935).

[21] Art. II, Liability Convention.

[22] Art. I(a), Liability Convention.

injury. Does that claim die with Mr. Smith? Or does the Liability Convention revive it? What of Mr. Smith's surviving family? May they bring an action for their injury resulting from the loss of his life?

Most states of the United States today resolve these questions by means of statutes ordinarily labeled wrongful death statutes and survival statutes.[23] Wrongful death statutes permit the surviving dependents to recover for their losses arising from the death, while a survival statute revives and vests in the decedent's estate the claim which the decedent could have asserted had he lived. A few states have so-called "hybrid statutes" which permit a measure of recovery equaling the combined recoveries permitted under both wrongful death and survival acts.

The Liability Convention does not tie the award for loss of life to the scheme of the particular state of the United States where a space object might impact. Thus a representative of the decedent's estate may recover as under a survival statute. Such an award could include compensation for the decedent's medical expenses between the time of injury and his death, for wages lost while unable to work as a result of the injury. for his funeral expenses, and perhaps for his pain and suffering. While the Liability Convention does not explicitly permit the recovery for the decedent's pain and suffering in the event those injuries subsequently cause his death, it does permit a recovery for "impairment of health" where a claimant suffers *personal injury*. Since impairment of *physical* health constitutes "personal injury," the Convention's reference to impairment of health must relate instead to impairment of emotional or mental health. This provision thus seems to permit recovery for the injured person's pain and suffering if he lives, and probably would allow a like recovery even when he subsequently dies of the injuries.

The injured person who subsequently dies from those injuries is not the only party who may suffer mental anguish as a result of his injury. A decedent's survivors also suffer grievous emotional anguish from their loss of a loved one. Once again the Liability Convention is silent concerning possible recovery for this type of "damage" or loss. Indeed, the Liability Convention does not address the broader question of whether a decedent's surviving family are proper claimants. Yet their loss is more immediate and real than that of the decedent's estate. If the deceased member was the wage earner for the family, then his demise may result in loss of family income, and the attendant financial support of his surviving spouse and children. Loss of inheritance is another element of injury suffered by the survivors. But for his death, the decedent may have left an estate to be inherited at his death by his family. Yet his untimely death will preclude this possible contribution.[24] The survivors may also have paid part or all of the decedent's medical expenses and even the funeral costs. Only if the survivors are

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[23] *See generally*, Dobbs, Remedies §8.2 (1973).

[24] *Id.* at § 8.3.

permitted to recover these several measures of loss will the Liability Convention serve its goal of moral right.

In the above hypothetical both Mr. Smith and Mr. Jones suffered personal injury, a compensable damage under the Liability Convention. As already noted, the losses produced by personal injury may include medical and hospital expenses, and lost wages or lost earning capacity. Where the injured person fully recovers from his injuries, his lost income will be incurred for an identifiable fixed period. In some situations, however, the injury may be so debilitating as to permanently limit or altogether preclude further ability to work. The Liability Convention does not speak to such future damages explicitly, but it does contain a statute of limitations which contemplates situations where the "full extent of the damage may not be known."[25] It is unclear whether such situations will result in an extension of the time for an initial claim or merely an extension *after* filing for discovering the full extent of prayed-for recovery. What is made clear, however, is that the Convention does permit recovery of damages not fully measurable at the time a claim is asserted. This is precisely the circumstance of future damages. Thus, if an injured person can establish that his injury permanently impairs his ability to work, the Liability Convention may permit recovery of such future damages.

As already stated, the Liability Convention includes impairment of emotional or mental health as a compensable injury. Still remaining is the practical problem of measuring such a recovery. As has been noted by a prominent scholar "it seems clear that the damages for pain and suffering are not compensation in any ordinary sense that they make the plaintiff whole or replace what has been lost, since damages are not pecuniary and since there is no market in pain and suffering by which the damages could be estimated."[26] How this recovery will be measured is no more definitely defined by the Liability Convention than that it be just and equitable. The Convention's second goal of restoring the injured person to the condition which would have existed without the injury is clearly inapplicable since a money award can never restore an earlier emotional state of mind nor eradicate the memory of suffering.

Many times a person who has suffered personal injury will be partially compensated for his loss by some independent source. Whether that source is private insurance, a state compensation scheme, or from a private benefactor, such compensation also serves to partially restore the injured person to the position occupied prior to the injury. The Liability Convention does not address that situation where some collateral source may partially accomplish this same remedial goal of the Convention. Should all such collateral benefits be subtracted from the claimant's recovery under the Convention? Would justice be served by subtracting insurance benefits which had been derived from a policy which had been paid for by the injured person? Or should only those collateral

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[25] Art. X, Liability Convention.

[26] *See,* Dobbs, Remedies 545 (1973).

benefits be subtracted which were rendered by some source to which the injured person had neither paid nor contributed?

There is a fourth remedial measure not mentioned in the Liability Convention. This measure, punitive damages, has been defined as "damages, other than compensatory or nominal damages, awarded against a person to punish him for his outrageous conduct"[27] or to deter similar acts by others. While it is difficult to conceive that a diplomatically negotiated compensatory award would ever contain an agreed element of punitive damages, it is possible that a claims commission might consider the appropriateness of such a remedy.

It is generally agreed that punitive damages are awarded due to the wrongdoing party's culpable state of mind. The requisite degree of cuplability is often described as "malice." Malice is in turn generally defined as evil intent or reckless disregard of the rights of the injured person. The Liability Convention does contemplate a situation where one State may act with the "intent to cause damage."[28] The Convention provides that a launching state will be exonerated from absolute liability in the event that it can establish that the damage caused by its space object was really the result of the claimant State's act or omission done with intent to cause damage. The Convention further provides, however, that no exoneration can be granted "in cases where the damage has resulted from activities conducted by a launching State which are not in conformity with the international law including, in particular, the Charter of the United Nations and the Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, including the Moon and Other Celestial Bodies."[29] If a claims commission were to refuse to exonerate a launching state for its failure to comply with such international law, that finding of failure would approach the standard of malice which is often sufficient to justify an award of punitive damages. If the claims commission further found that the purposes of punitive damages to punish or deter would be served, then that commission might well designate an award that included a measure of punitive damages. While such a measure might constitute a "damage" arising from the launching State's activity, it would certainly go beyond the remedial goal of restoration.[30]

The above stated hypothetical found Messrs. Smith and Jones standing on Mr. Smith's realty at the time of impact. Assume now that situated upon that land was a

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[27] *See,* Restatement of Torts §908 (1939).

[28] Art. VI ¶ 1, Liability Convention.

[29] Art. VI, ¶ 2, Liability Convention.

[30] *See generally,* Ghiardi, The Case Against Punitive Damages, 8 Forum 411 (1972); Hodgin and Veitch, Punitive Damages Reassessed, 21 Internat. & Comp. Law Quart. 119 (1972); Exemplary Damages in the Law of Torts, 70 Harv. L. Rev. 517 (1957); Prosser, Handbook of the Law of Torts 13 (4th ed. 1971); McCormick, Damages Chp. 10 (1935); 22 Am. Jur. 2d, Damages §261 (1965).

manufacturing business owned by Mr. Smith. Within that structure were various items of personalty which were integral to that business. If the falling space object injured or destroyed the structure containing the manufacturing business and contaminated the land so as to limit or preclude any future use, Mr. Smith will have suffered substantial additional injuries. The Liability Convention allows recovery for "loss of or damage to property" without limiting the property concept to either realty or personalty. Thus, destruction or injury to even intangible personalty would produce a compensable loss under the Convention. Again not stated is the measure of any such compensation.

If the total property is only injured, can the property owner recover an amount equal to the diminution of that property's market value, the cost of restoring the property to its prior condition, or may he recover the larger of the two amounts regardless of whether he in fact repairs the injury?[31] Can he also recover for the loss of use of this property?[32] That loss would occur during the period necessary to repair.

How will such loss be measured? Will that measurement be limited to the market rental value of the property, to the owner's particular use value, to the lesser of the two, or the greater?[33] May the owner also recover interest on the loss?[34] If so, will the interest accrue from the date of injury or some later date?[35]

If the property is totally destroyed, similar measuring questions are presented.[36] Does the owner recover the property's value as measured by his purchase price, by its market value on the date of destruction, or on the date of award? Can be also recover for loss of use? Even if lost use or profits? is compensable, for what period are they recovered? That is, must the owner attempt to minimize this loss by promptly starting

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[31]"In addition to damages for the diminution of the value of the subject matter or other similar element of damages, the plaintiff is entitled to recover for any loss of which the defendant's act is the legal cause, either because the plaintiff is unable to use the subject matter until it is repaired or replaced or otherwise." Restatement of Torts §928, comments *(b)* (1939).

[32]Dobbs, Remedies §§5.3 (loss of use of realty) and 5.11 (loss of use of personal property) (1973).

[33] *Id.* at 333.

[34] *Id.* at §3.5.

[35] *Id.*

[36]"When a chattel is damaged beyond repair and is treated as destroyed, the usual measure of damage is the market value of the chattel at the time and place of destruction with adjustments for salvage value." *Id.* at 375. (footnotes ommited). *See also, Id.* at 379 (measure of damages where personal property is damaged rather than destroyed). *See generally,* McCormick, Damages, Chp. 19 (1935) (injury to interests in personal property).

anew his destroyed business?[37] What if he lacks sufficient capital to initiate such an endeavor until compensated by the launching State?[38] Is his award then limited to that period when a sufficiently capitalized hypothetical owner could have begun this business afresh?

The purpose of this paper has been to identify some of the questions which may arise when first applying the Liability Convention's remedial scheme to an actual injury. While few answers are proffered to the several questions raised, it must be remembered that any answers to the Convention's remedy questions should be arrived at through application of its stated remedial goals. To do less would defeat the obvious intention of the drafters to do justice for the *injured persons* when fashioning individual awards.

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[37]"(1) Except as stated in Subdivision (2), a person injured by the tort of another is not entitled to recover damages for such harm as he could have avoided by the use of due care after the commission of the tort.

(2) A person is not prevented from recovering damages for a particular harm resulting from a tort if the tortfeasor intended such harm or adverted to it and was recklessly disregardful of it, unless the injured person with knowledge of the danger of such harm intentionally or heedlessly failed to protect his own interests." Restatement of Torts § 918 (1939). At least one court has held that the injured party may not recover for preventable loss that the injured party could have prevented but for his financial inability to do so. Brandon v. Capital Transit Co., 71 A. 2d 621 (Mun. Ct. App. D.C. 1950).

[38]For one court's holding that the injured party's financial inability to mitigate does not preclude recovery for otherwise preventable loss, *see,* Valencia v. Shell Oil Co., 23 Cal. 2d 840, 140 P. 2d 558 (1944).